

Design and Testing of a Deployment Mechanism for NASA's 1,653-m² Solar Cruiser Sail

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Abstract

NASA Marshall Space Flight Center (MSFC), in collaboration with Redwire and NeXolve, is advancing the design of a 1,653-tracm² Solar Sail System (SSS) for the Solar Cruiser mission. Since 2019, the program has been developing key components, including: the Sail Deployment Mechanism (SDM), high strain composite Triangular Rollable and Collapsible (TRAC) booms, and the ~ 413 m² thin film sail quadrants. In 2023 and 2024 validation testing was conducted to advance the SSS design to technology readiness level (TRL) 6 maturity. This testing demonstrated the challenges of using compliant mechanisms on such a large scale, and how the mechanism design and processes must be customized to enable the high strain composite structures to function properly. This paper describes the solutions the team found through the development and testing of a large-scale compliant structure which are necessary to support advanced mission concepts such as solar sailing missions.

Introduction

The SDM and TRAC booms [1], both developed and manufactured at Redwire in Longmont, Colorado, act as the deployment mechanism and support structure for the 1,653 m² Solar Cruiser solar sail (*Figure 1*), destined to orbit Lagrange point 1 and could support heliophysics studies along with other missions [2]. The TRAC booms are carbon fiber reinforced plastic (CFRP) with a laminate architecture referred to high strain composites. Redwire's Longmont facility (formerly Roccor) has specialized in high strain composite mechanisms for over a decade; each application comes with its challenges. For TRAC booms, these challenges come in the form of undesired buckling and delamination of the bonded edge during the stowing or deploying process. The latest laminate architecture designed for the Solar Cruiser mission proved to be resistant to both of these phenomena in early testing. After 4 years of development Redwire constructed the TRL 6 prototype to demonstrate the design's capability to survive environmental testing and deploy a full-scale sail quadrant.



Figure 1. Solar Cruiser TRL 6 prototype following single quadrant deployment testing.

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Through two design/testing cycles, TRL 5 and TRL 6, the engineering team found and applied lessons learned to reach the results presented in this paper. This paper will primarily focus on the development of the mechanisms interfacing with the TRAC booms and the inherent challenges in testing compliant mechanisms for spacecraft on this large scale.

Background

Solar Sail Engineering Best Practices

The idea of solar sailing dates to the 1920's [3], but methods for engineering practical sail systems have only begun to emerge within the past 20 years. In 1999 the German Aerospace Center (DLR) proposed a now commonly accepted structural architecture for solar sails that includes four flexible booms co-rolled on a common reel and deployed in unison to tension four interconnected sail membrane quadrants [4]. This type of sail architecture was first flown in 2011 by NASA with the NanoSail-D mission [5].

At the 2016 Aerospace Mechanisms Symposium, engineers from NASA Marshall Spaceflight Center presented an 86 m² version of the 10 m² NanoSail-D2 sail system then under development for the planned NEA Scout mission [6]. That paper focused on the obstacles of scaling up the four-boom deployment mechanism to accommodate a larger deployed sail, and the lessons learned from a thorough development process. Design and flight-qualification of the NEA Scout system proved to require the development of approximately half a dozen prototypes to identify unforeseen problems, advance solutions, and build confidence in the final design product. The NEA Scout deployer utilized TRAC booms made of Elgiloy steel, which were prone to blooming and greater thermal deflection in comparison to composite alternatives. Meanwhile, Redwire developed a family of compact deployment mechanisms for high-strain composite booms [7] and advanced manufacturing and design best-practices for large-scale Triangular Rollable And Collapsible (TRAC) booms and their deployment mechanisms [8] to support NASA's quest for practical large-scale solar sail missions.

Challenges in Developing the Solar Cruiser Sail System

In 2020, A team led by NASA's Marshall Space Flight Center and including Redwire and NeXolve was funded to develop the Solar Cruiser mission to mature solar sail technology for future mission use in the 5–15-year timeframe [2]. The Solar Cruiser sail is the largest yet designed for flight with a 1,653 m², 2.5-micron thick sail membrane deployed by four 29.5-m composite TRAC booms (see *Figure 2*). In comparison to the 86 m² NEA Scout sail deployment mechanism, Solar Cruiser's deployment mechanism measures over ten times larger, while sharing many of the same engineering and development challenges (see *Figure 3*).

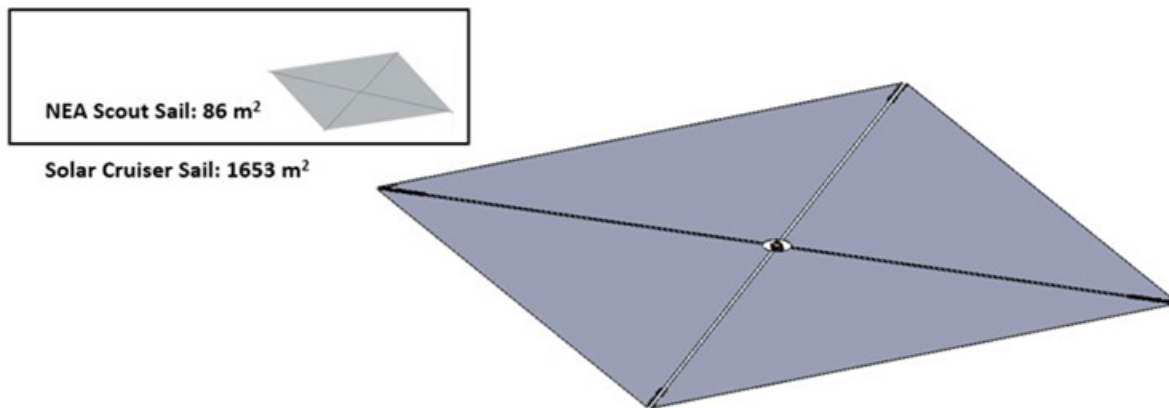


Figure 2. Solar Cruiser sail system compared with NEA Scout system, in deployed states.

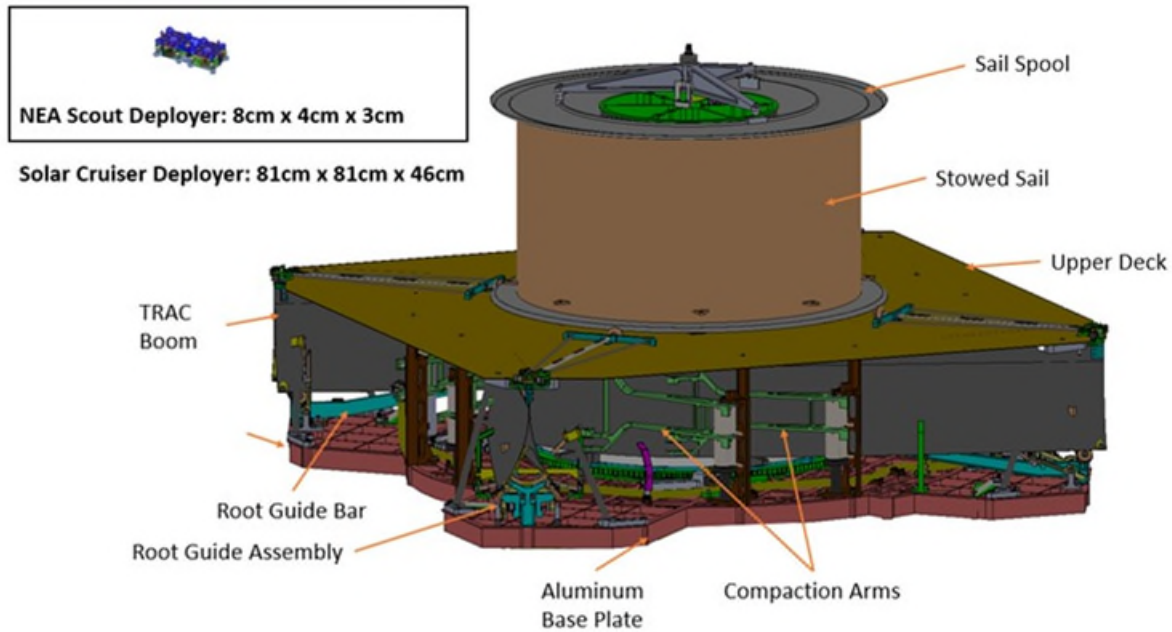


Figure 3. Solar Cruiser sail system compared with NEA Scout system in stowed states.

Scaling the sail deployment mechanism starts with scaling the deployed TRAC boom to accommodate longer length, higher stiffness, and increased stability requirements for the larger sail application. While attention must be given to structural scaling for deployed performance, equal attention must be given to ensuring the boom design provides adequate furling performance. Specifically, inner flange buckling of TRAC booms (see Figure 4) is a complex failure mechanism that can develop during boom furling and lead to packing inefficiencies and boom failure [8]. Inner flange buckling was a significant issue for both the NEA Scout sail deployment mechanism development [6] and the Solar Cruiser deployment mechanism development discussed here.

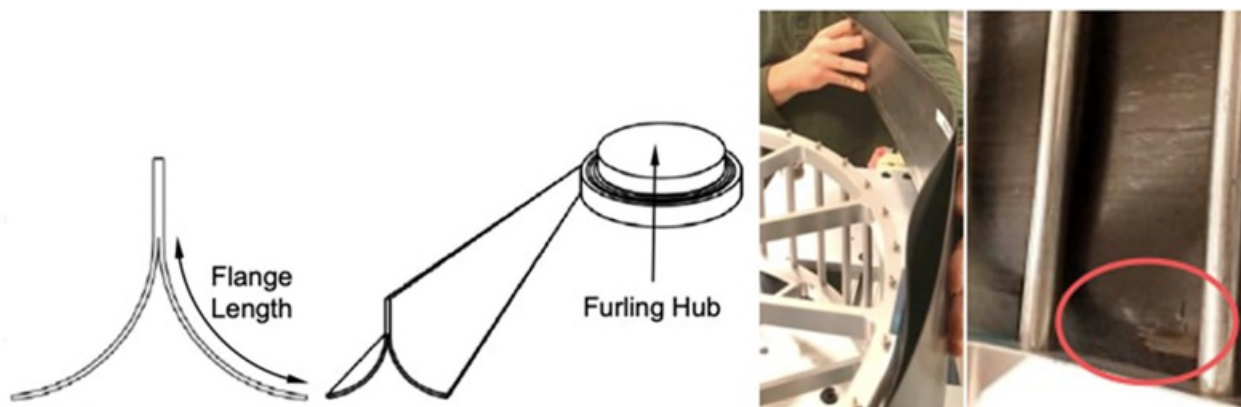


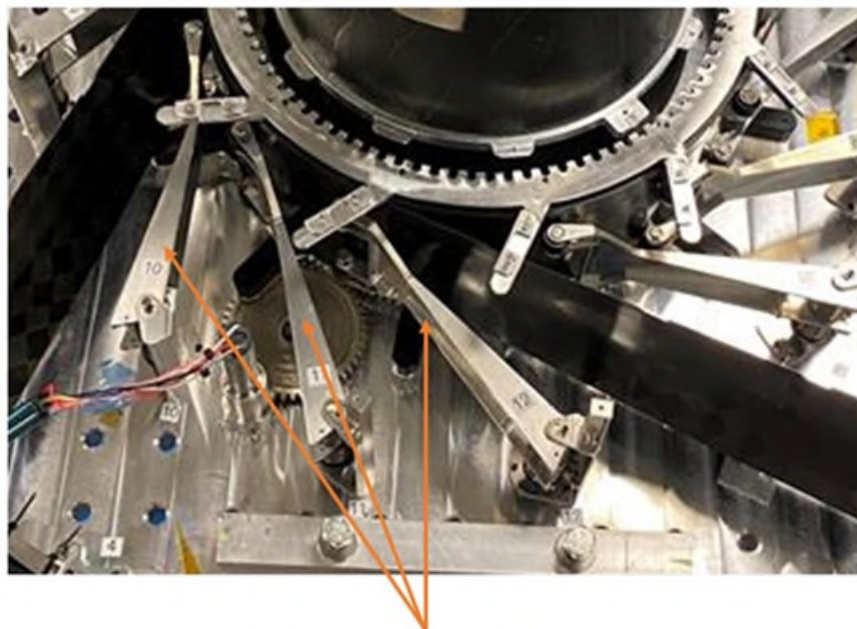
Figure 4. TRAC boom cross section (left), furling configuration (center-left), inner flange buckling mode (center-right), and material fracture failure mode enclosed in red circle (right).

Early during the Solar Cruiser program, an extensive engineering study was performed by Redwire to optimize the TRAC boom design to best-meet the packaging and deployed structural performance requirements of the mission [9]. This development program encompassed testing of a series of full-scale and sub-scale test articles in combination with an extensive finite element analysis effort to characterize packaging and deployed structural performance. In parallel with this effort, a series of EDUs were developed and tested for the system culminating in a first full-scale deployment test of an initial (TRL 5)

system including two TRAC booms and a single sail quadrant in 2022 [10] and a second full-scale deployment test of a flight-like (TRL 6) system in 2023-2024. The present paper summarizes lessons learned from this engineering development program.

Brassboard Prototype and TRL 5 Testing

The Brassboard prototype allowed for rapid testing and design iteration on the Solar Cruiser scale. The prototype allowed for functional testing of the mechanism while not meeting all flight requirements such as mass budget and thermal resilience. Full stowing efficiency, simply defined as a lack of inner flange buckles on multiple wrapped TRAC booms, was a goal for the new laminate architecture and Brassboard prototype. Notably, the design includes compaction arms and keys to contain the TRAC booms in their stowed wraps. The compaction arms have torsion springs which provide inward force through rollers to the stowed TRAC booms' outer circumference (see Figure 5). This compaction force prevents inner flange buckling during the furling and deploying processes.



Compaction Arms

Figure 5. Compaction arms on Brassboard prototype.

The keys are fixed to the top of the boom hub (see Figure 6), and interface with notches [11] cut into the web of the boom. The notches are cut early in the assembly process to allow for cleaning processes to take place. The deployment is driven by a single motor rotating the boom hub. The notches act as hard stops for the keys preventing blooming between wraps during deployment and driving the booms outward.

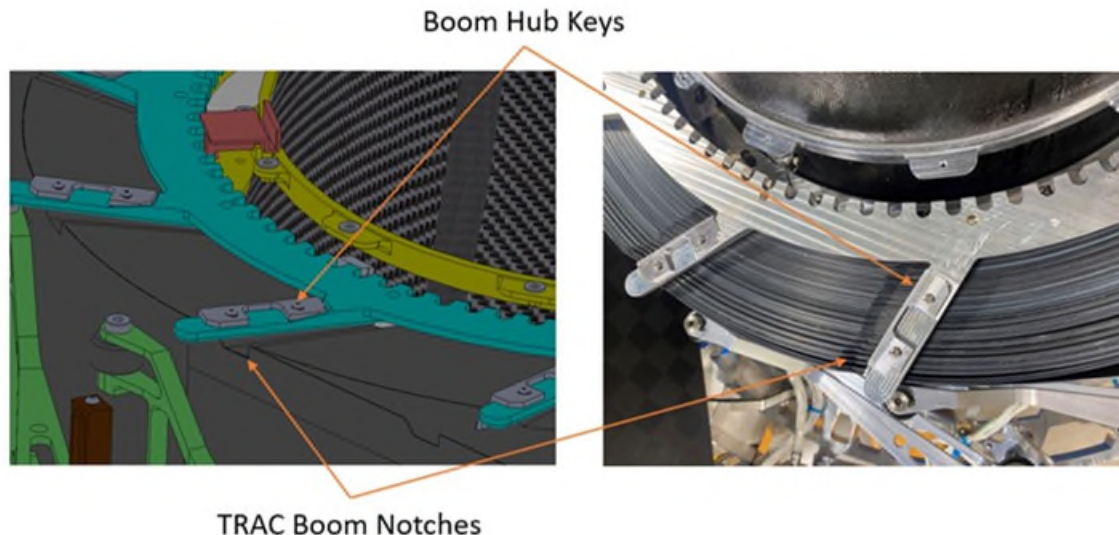


Figure 6. Keys and notches shown in the Solar Cruiser CAD (left) and actual prototype (right).

This brassboard prototype SDM was functionally tested in two forms: 1) with four subscale booms and 2) with two full length booms. Both tests yielded a better understanding of the capability of the sail deployment system and the behavior of the CP-1 sail material as it unfolds in ground testing. A detailed summary of these tests is included in “Test of a Full-Scale Quadrant for the 1,653 m² Solar Cruiser Sail” [10].

The first brassboard prototype test demonstrated that four 7.5-meter TRAC booms could be stowed effectively with no inner flange buckles or delamination and deploy four 1/16th scale area sail quadrants wrapped on a single hub simultaneously. During this test the booms were deployed above the sail quadrants to avoid any snagging on the notches as shown in Figure 7.

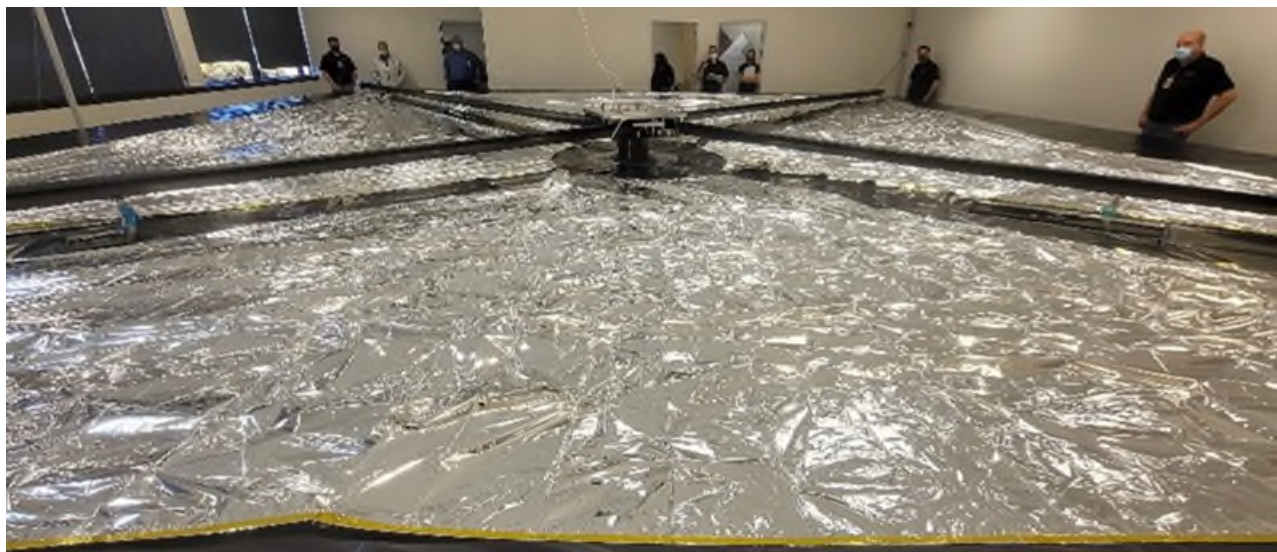


Figure 7. Brassboard Prototype deploying four quarter length booms and four 1/16th scale area sail quadrants.

The second brassboard prototype test demonstrated that two 30-meter TRAC booms could be stowed effectively with no inner flange buckles or delamination, and the SDM could deploy a full-scale sail quadrant as shown in Figure 8. During this test the sail quadrant was deployed below the booms and along a layer of anti-static film to minimize friction with the floor and un-realistic deployment resistance forces.

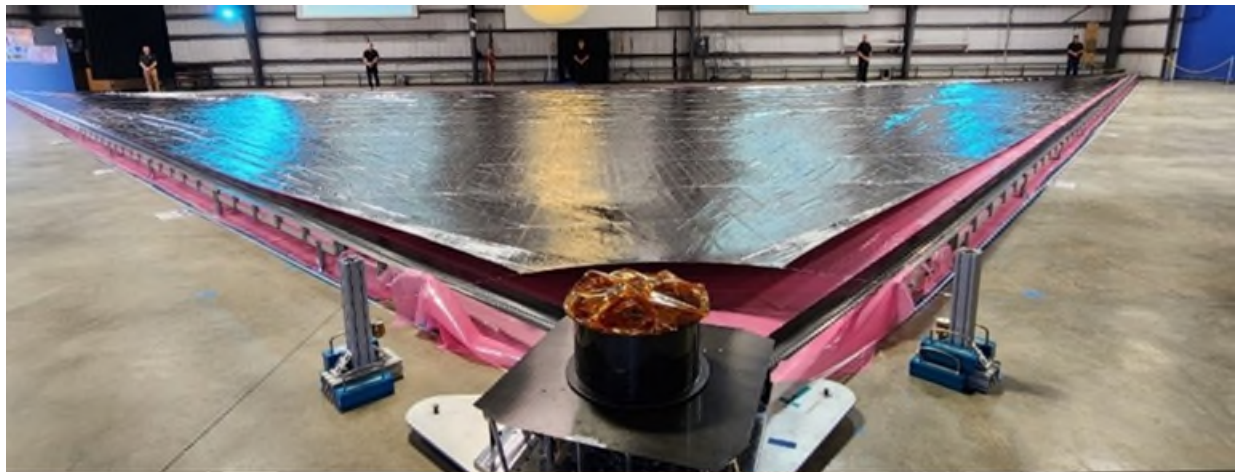


Figure 8. Brassboard prototype after deploying a single full scale sail quadrant with two full length booms.

Challenges with Ground Testing

Ground testing the Solar Cruiser deployment system in gravity presents many challenges, as the mechanism is designed to function in a microgravity environment. For the full-scale test, a single sail quadrant was deployed instead of the four quadrants in the flight design, to limit the cost and complexity. Finding a facility large enough to deploy a single quadrant was a challenge in early planning; much more difficult to find one for more quadrants. Access to the deployer and monitoring the sail for snagging were concerns for the deployment test, which were simplified by only deploying the single quadrant. The ground support equipment (GSE) roller and track assemblies supported the booms against the vertical and lateral loads experienced due to gravity and the absence of adjacent sail quadrants. The deployment area for the sail was covered in an anti-static low friction material to prevent snagging. The sail was deployed on the ground to allow for sail snag monitoring through relatively higher vantage points and to remove the cost of manufacturing a massive table surface.

In first assembly, two TRAC booms of the Brassboard deployer were supported by lower friction rollers than the GSE rollers used in the deployment test. This resulted in a newly observed phenomena known as notch migration. Due to the slightly greater tension put on the booms in this subsequent furling, the layers wrapped tighter on each other, shifting their relative notch locations. This only became an issue in the outer wraps which caused a slight interference between the key and notch. From this the team learned that the TRAC booms must be initially furled in the same loading conditions as future stowing procedures following deployment tests.

Solar Cruiser TRL 6 Prototype Assembly and Testing

To achieve a TRL 6 designation, the design team within Redwire built on the success of the Brassboard prototype, with a lightweighted design featuring flight materials and coatings. Solar sail performance requires a high area to mass ratio to achieve the necessary thrust performance. The updated design still used the same boom interfacing mechanisms developed on the Brassboard, with compaction arms, keys and notches. The TRL 6 prototype satisfied the flight configuration of four full length TRAC booms and only eight compaction arms for lightweighting. Once the assembly was complete, the solar sail system underwent vibration and thermal vacuum (TVAC) testing (see Figure 9). Intermittently throughout these tests, the launch restraints were released, and booms partially deployed to show functionality in environmental testing. Following these tests, a full-scale sail quadrant deployment test was performed, with two booms fully deployed outward and two spooled on GSE spools in January of 2024 (see Figure 1).

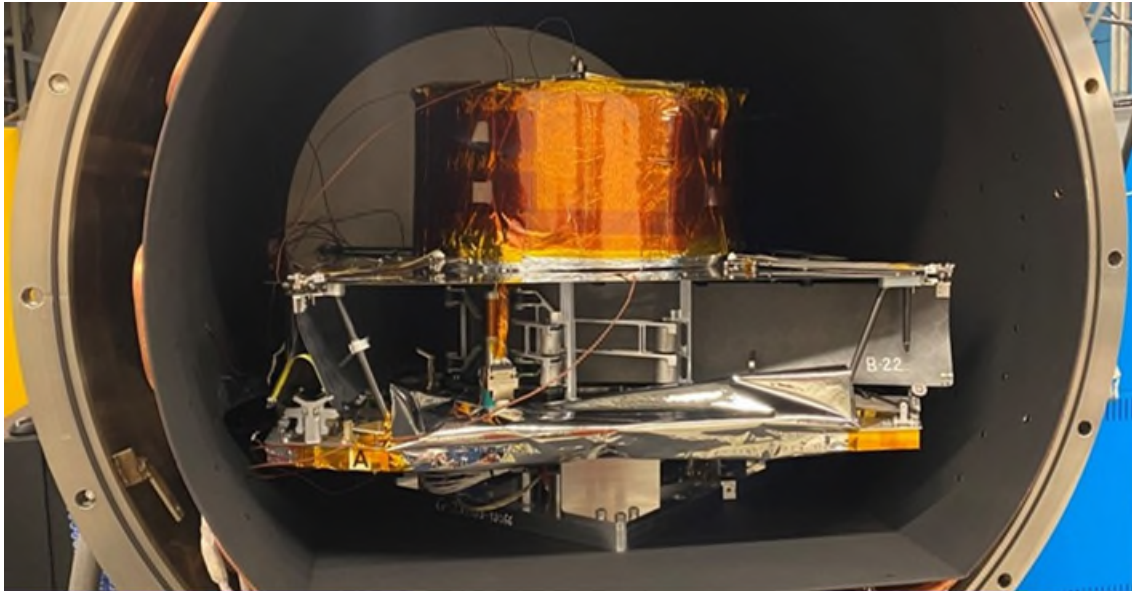


Figure 9. TRL 6 prototype in TVAC chamber before testing.

Complications Due to Inner Flange Buckling

Early in the assembly process, the four full-length TRAC booms must be stowed onto the deployer. This was foreseen to be challenging with the eight-compaction arm design, so four additional GSE compaction rollers were added to the system, which aided in the stowing process. After roughly a third of the boom length had been stowed, inner flange buckling began to occur. These buckles can be mitigated by manually applying pressure to the buckle location or reversing the motor direction; these mitigation strategies can also increase the buckle's amplitude or shift them farther in the wrap (see lefthand photo in Figure 10). The stowing process had been slowed down immensely, with buckles continuing to form until the end of stowing, increasing in frequency towards the outer wraps. Due to the successful furling of the booms on the Brassboard prototype, the inner flange buckling was not foreseen as an issue that would cause deployment failure on the TRL 6 prototype. As full stowing efficiency was not a project requirement, the project team agreed to accept inner flange buckling due to assembly timeline constraints. Therefore, notches were cut into the booms in this stowed configuration (see righthand photo in Figure 10).

Note that during testing a revised boom-stowage procedure was developed that completely eliminates the incidence of inner-flange buckling. This revised stowage procedure and implications for future work will be discussed later, but at this early juncture in the program a certain amount of inner flange buckling was accepted along with a degree of detrimental functional performance of the system due to the incidence of inner flange buckling. For example, the decreased stowing efficiency due to inner flange buckling was measured in roughly 1 cm asymmetrically added to the outer diameter of the stowed booms. The added diameter to the stowed booms on the hub caused an aluminum root guide bar positioned between the flanges of the TRAC boom, to bend outward beyond its intended travel. A small bearing on the end tip of the bar failed in early testing, breaking into pieces from the additional radial load in this deflected position. This bearing was repaired before final assembly, and not damaged again, but was an early indication to the importance of full stowing efficiency. The angle of the boom at its tangent point on the stowed wrap relative to the guide frame on the edge of the deployer changes as booms are furled onto the hub. The guide frame has rollers which provide downward force to the top web of the booms, the width of this roller is also based on the assumption of full stowing efficiency. Towards the end of the stowing process, the boom web tended to move to the side of the roller, due to the boom tangency angle changing beyond what was expected of the design. The assumption in design of full stowing efficiency led to these complications and has been noted as a potential tolerancing improvement on future designs.

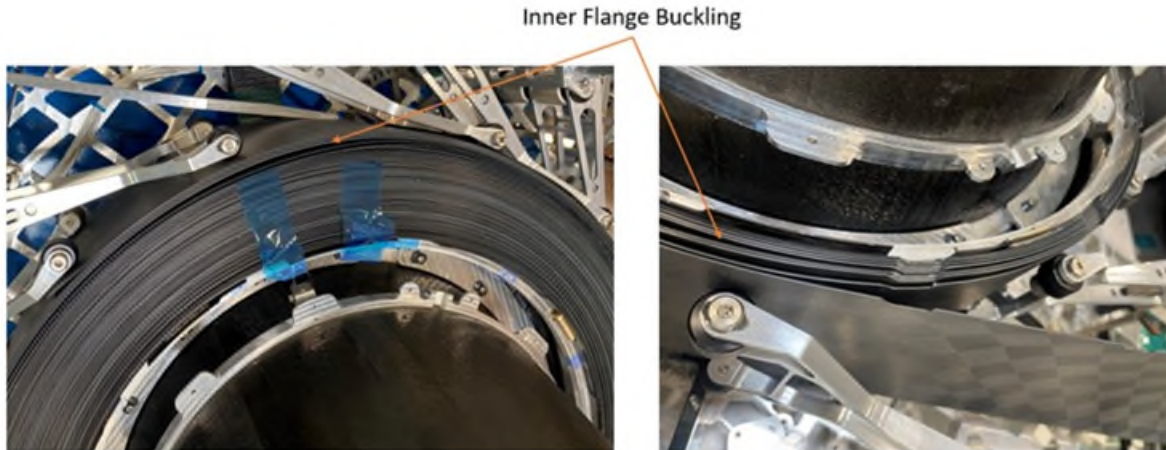


Figure 10. Inner flange buckling on TRL 6 prototype before (left) and after (right) notching process.

From the lessons learned on the Brassboard testing, the four full length TRAC booms were initially stowed on the same GSE intended for TRL 6 testing. This was intended to prevent the notch migration observed from changing boom support conditions. However, inner flange buckles form somewhat randomly during the stowing process, so there is no guarantee that boom notches will align properly after repeated deployment and stowage cycles. Indeed, variations in the formation of inner flange buckles were observed on the final deployment and stowage cycle for the SDM and the four TRAC booms prior to integration with the sail quadrant for final system deployment testing (see Figure 11). For this reason, a second notch cutting process was conducted in the final assembly steps leading up to testing of the complete system. Modifying these notches introduced foreign object debris (FOD) to the assembly, an acceptable level for ground testing but undesired for flight missions with contamination requirements. A second notch cutting processes would not be taken for a flight unit, however they were necessary to continue with planned ground testing.



Figure 11. Notch migration due to inner flange buckling, prior to second notch cutting process.

In the assembly process, a single deployed TRAC boom was measured and cut to the designated length. The other three booms were cut to length relative to points on the deployer, after a partial stowing procedure. The inner flange buckling was present in this process, resulting in the booms not fully stowing consistently in subsequent stows. A single boom would reach its fully stowed position while the other three remained slightly deployed to varied lengths. The greatest length being 1.3 cm from full stow, inhibiting launch restraint features in the system. To properly dock for vibration testing, the booms were manually pushed into position. This resulted in blooming of the outer wrap of the repositioned booms between compaction arm rollers. This blooming was shaken out in vibration testing, causing the booms to prematurely return to their not fully stowed states. In a flight configuration this would result in a premature tearing of the sail's launch restraint membrane.

Small design changes will be able to mitigate the complications mentioned above, however it is apparent that achieving TRAC boom furling without inner flange buckling is essential. Furthermore, it is essential to provide the time and attention to achieve full stowing efficiency before the booms are notched or cut to length. The decision to move forward with the assembly despite having inner flange buckles resulted in many unforeseen complications in the remainder of testing. At the time of this decision, it was unknown if the four TRAC booms could be stowed with full efficiency at the current scale. Following the successful deployment test, this uncertainty was reduced with the team accomplishing a full efficiency stow without inner flange buckles.

Post Test Stowing – Elimination of Inner Flange Buckles

Following the full-scale single quadrant deployment test of the TRL 6 prototype shown in *Figure 1*, the system was left in its deployed state for sail measurements to be taken. Once the sail was folded and removed, the team cautiously began the stowing process, knowing inner flange buckling was likely to occur. With careful attention, and moderate manual adjustments, the team was able to effectively stow the four full length booms on the sail deployment mechanism without any inner flange buckles. Some new techniques were utilized to adjust the height of the booms, deployer and GSE relative to one another. Even after a successful stow without inner flange buckles, the combined thickness of all the wrapped booms was 5.9 cm, where in the CAD it was represented with a thickness of 5.2 cm. This difference in thickness exemplifies an assumption of stowing efficiency that can be better accounted for in the deployer's design. As the notches were originally cut and recut with inner flange buckles present, the notches were misaligned. Improved processes, GSE and component level design modifications will improve the repeatability of stowing without inner flange buckles and improve the mechanism's resiliency to losses in stowing efficiency.

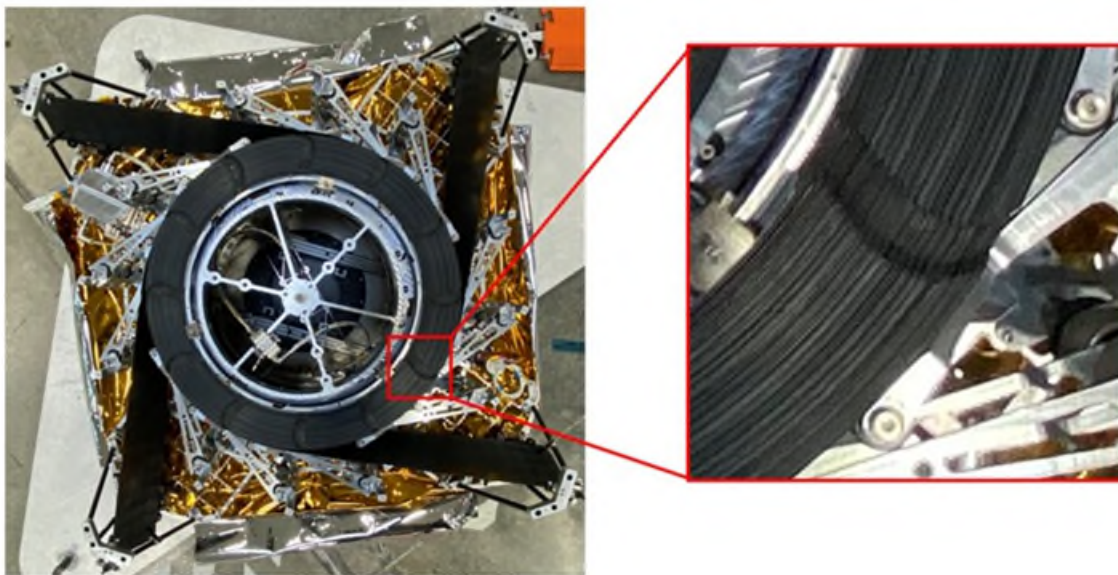


Figure 12. Successful full efficiency stowing with no inner flange buckling (left) and observation of the resulting migration of notches (right).

Conclusion

Through two design and testing cycles, TRL 5 and TRL 6, the engineering team at Redwire has been able to demonstrate the capabilities of the Solar Cruiser sail deployment mechanism, identifying challenges and finding solutions throughout. The design of the laminate architecture of the TRAC booms and interfacing mechanisms enabled a massive increase in scale of the solar sail deployer, when compared to previous flight designs. The success of the TRL 5 testing led to the flight capable design of the TRL 6 prototype, which came with the unexpected challenge of inner flange buckling. The loss in stowing efficiency coupled with a design assumption of full stowing efficiency, led to various losses in performance during testing. The engineering team's follow-on process improvements resulted in a full stow without inner flange buckles, validating the TRL 6 design and revealing design adjustments to enable a flight capable sail deployment mechanism.

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